



AoPS has prepared students for the most prestigious high school competitions since 1993. Over the last decade, most winners of the USA Mathematical Olympiad (USAMO) have been AoPS students, including 15 USAMO winners in 2020, as well as the US team members that won the International Math Olympiad (IMO) in 2015, 2016, 2018, and 2019. In this course, students study strategies crucial for success on the American Mathematics Competitions 10 and 12 (AMC 10 and 12), the American Invitational Math Exam (AIME), the Harvard-MIT Math Tournament, and other major high school contests.

Students should either have completed an Algebra 2 course or be currently enrolled in one to be successful in this course.

Note for students who have taken this course: We alternate between two different versions of this course from one year to the next. The two versions are roughly the same difficulty, but include different lessons, practice contests, and homework. Therefore, students taking this course for two consecutive years will see different material in the second year than in the first.

Homework and exams: Students should expect to spend 1-3 hours on homework every week. Homework will be assigned and completed through the class website, where students receive immediate feedback on their work. Students will take four in-class exams, structured as practice math contests, throughout the course.

Book: *The Art of Problem Solving, Volume 2: and Beyond* by Sandor Lehoczky and Richard Rusczyk

Time Commitment: This 36-week course includes 105 minutes in-class and 1-3 hours of homework each week, corresponding to a full year course.

Version A Content:

Week	Topic
1	Systems of Equations
2	Logarithms
3	Polynomials
4	Sequences and Series
5	Functions and Operations
6	Trigonometric Identities
7	Complex Numbers
8	Special Factorizations
9	Mock Contest A
10	Triangle Basics
11	Pythagorean Theorem
12	Special Right Triangles
13	Polygons
14	Circles A: The Basics
15	Circles B: and Beyond
16	Funky Areas
17	3D Geometry
18	Mock Contest B

Week	Topic
19	Counting Basics A
20	Counting Basics B
21	Casework
22	Indistinguishable Objects
23	Principle of Inclusion and Exclusion
24	Combinatorial Identities
25	Correspondences
26	Counting on the Plane
27	Mock Contest C
28	Primes and Composites
29	Greatest Common Divisor
30	Modular Arithmetic A: The Basics
31	Modular Arithmetic B: and Beyond
32	Diophantine Equations
33	Bases
34	Combinatorial Expressions!
35	Digits
36	Mock Contest D

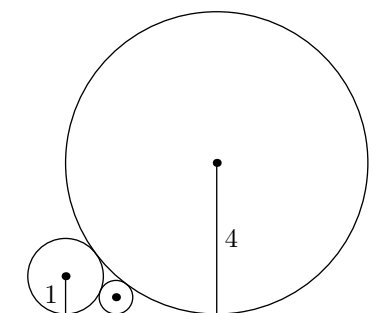
Version B Content:

Week	Topic
1	Equations and Systems
2	Polynomials
3	Sequences and Series
4	Trigonometric Identities
5	Rates
6	Quadratics
7	Optimization
8	Special Functions
9	Mock Contest A
10	Triangle Basics
11	Right Triangles
12	Quadrilaterals
13	Circles A
14	Circles B
15	Trigonometry
16	Coordinate Geometry
17	3D Geometry
18	Mock Contest B

Week	Topic
19	Basics / Correcting for Overcounting A
20	Correcting for Overcounting B
21	Casework Counting
22	Constructive Counting
23	Distributions
24	Probability A
25	Probability B
26	Recursion
27	Mock Contest C
28	Divisors and Multiples
29	Primes and Composites
30	Perfect Squares
31	Factors of an Integer
32	GCD and LCM
33	Divisibility Tests
34	Digits and Number Systems
35	Diophantine Equations
36	Mock Contest D

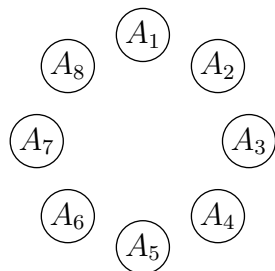
Sample Problems:

- The fifth and eighth terms of a geometric sequence of real numbers are $7!$ and $8!$ respectively. What is the first term?
Source: 2009 AMC 12B #12
- A circle with a radius of 1 and a circle with a radius of 4 are externally tangent. A third circle is tangent to the first two and to one of their common external tangents as shown. Find the radius of the third circle.



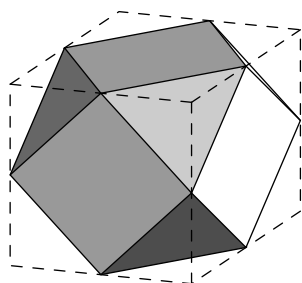
Source: 2001 AMC 12 #18

- Eight people are sitting around a circular table, each holding a fair coin. All eight people flip their coins and those who flip heads stand while those who flip tails remain seated. What is the probability that no two adjacent people will stand?



Source: 2015 AMC 10A #22

- If a and b are positive integers such that $\frac{1}{2} + \frac{1}{a} = \frac{1}{3} + \frac{1}{b}$, what is the sum of all possible values of a ?
Source: 2014 MathCounts State Team #9
- Everyday at school, Jo climbs a flight of 8 stairs. Jo can take the stairs 1, 2, or 3 at a time. For example, Jo could climb 3, then 1, then 2, and finally 2 more. In how many ways can Jo climb the stairs?
- How many 7-digit palindromes are there?
- For each vertex of a solid cube, consider the tetrahedron determined by the vertex and the midpoints of the three edges that meet at that vertex. The portion of the cube that remains when these eight tetrahedra are cut away is called a *cuboctahedron*. Find the ratio of the volume of the cuboctahedron to the volume of the original cube.



Source: 1992 AHSME, #19

- Let A , M , and C be nonnegative integers such that $A + M + C = 12$. What is the maximum value of $A \cdot M \cdot C + A \cdot M + M \cdot C + A \cdot C$?
Source: 2000 AMC 12 #12